LEVEL 20 | WHEATSTONE BRIDGE BALANCE

Unity prefab construction / wiring / ID declarations / evaluation / validation

Goal: replace the incorrect 220 ohm RIGHT TOP resistor with 330 ohm; balance the left and right 330/470 ohm dividers so the LCD meter reads less than 0.05 V between midpoints.

1. Electrical layout and connection sequence

Important: the dashed connection between MID A and MID B is a **high-impedance LCD voltage measurement**, never a SparkSingleWire. Wire each leg between the same positive and negative supply rails.

```
5 V (+) -- Power Switch --+-- LEFT TOP 330 ohm -- MID A -- LEFT BOTTOM 470 ohm --+ | | +-- RIGHT TOP
330 ohm -- MID B -- RIGHT BOTTOM 470 ohm -+-- 5 V (-) LCD Terminal A -> MID A LCD Terminal B -> MID B
```

Use distinct terminal connections at each midpoint. For a replaceable part, use the installed resistor terminal objects, not a stale prefab reference.

2. Assemble the original Level_20_Layout prefab

1. Open **Level_20_Layout** in Prefab Mode. Keep the root SparkLevelLayout, SparkLevelSceneBindings and SparkLevelWorkstationBindingBridge.

2. Create 4 **unique Workstation SparkShelfSlot** objects: Register_Slot_L_TOP, Register_Slot_L_BOTTOM, Register_Slot_R_TOP, Register_Slot_R_BOTTOM. Set category Resistor and Accept Any Category OFF.

3. Author the initial LEFT TOP 330, LEFT BOTTOM 470, RIGHT TOP 220 (deliberately wrong), RIGHT BOTTOM 470 components. Ensure each starting SparkShelfItem occupies its actual workstation (CurrentSlot / Occupant / InstalledItem).

4. Ensure the correct replacement 330 ohm part is accessible in persistent shelf storage; power must be off before removing any wired part.

5. Connect each series leg and common rails; wire the LCD A/B terminals to separate MID A and MID B test points. Do not connect MID A and MID B together as a conducting cable.

6. Save the original prefab outside Play Mode; verify the layout loader uses Level_20_Layout and the new Level_20.asset.

3. Original ScriptableObject asset

Import the companion **Level_20_Wheatstone_Bridge_Corrected.txt** via your Project Spark editable-SO importer, while selecting the ORIGINAL Level_20.asset. Asset GUID, path, serialized keys and markers are preserved.

WORKSTATION IDS AND TARGET RULES

4. Workstation bridge (four unique roles)

Slot ID	Component ID	Terminal bindings
Register_Slot_L_TOP	R_LTOP_Component	R_LTOP_Input / R_LTOP_Output
Register_Slot_L_BOTTOM	R_LBOTTOM_Component	R_LBOTTOM_Input / R_LBOTTOM_Output
Register_Slot_R_TOP	R_RTOP_Component	R_RTOP_Input / R_RTOP_Output
Register_Slot_R_BOTTOM	R_RBOTTOM_Component	R_RBOTTOM_Input / R_RBOTTOM_Output

In SparkLevelSceneBindings declare the **four component IDs** and **eight terminal IDs** above. They are declarations only; the bridge must populate live references.

Binding strategy: either configure those IDs on each SparkShelfSlot with Use Smart Slots ON, **or** configure four legacy workstation rules with Use Smart Slots OFF. Smart slot definitions override legacy entries with identical SlotIds.

5. Level target settings (already in corrected TXT)

Objective	Target pair / component	Acceptable range
Bridge balance (voltage)	R_LBOTTOM_Input vs R_RBOTTOM_Input	0 - 0.05 V
LEFT midpoint voltage	R_LBOTTOM_Input vs R_LBOTTOM_Output	2.80 - 3.10 V
LEFT branch current	R_LTOP_Component	5.8 - 6.7 mA
RIGHT branch current	R_RTOP_Component	5.8 - 6.7 mA
Total source current	PowerSupply_01	11.8 - 13.2 mA

Evaluator detail: Your current VoltageRange implementation reads the terminal pair first and compares absolute voltage difference. For bridge balance, TargetComponentId is also set to R_LBOTTOM_Component so the target remains configured with current IsConfigured() implementations that require a component or single terminal ID.

6. Expected readings

State	MID A (L)	MID B (R)	Bridge delta	Total current
Starting wrong R_TOP=220 ohm	2.9375 V	3.4058 V	0.4683 V	13.496 mA
Corrected R_TOP=330 ohm	2.9375 V	2.9375 V	~0.000 V	12.500 mA

The resistor branch currents when balanced are approximately 6.25 mA each. Slight differences are expected due to simulated switch and wire resistances.

PLAY MODE VALIDATION AND TROUBLESHOOTING

7. Step-by-step test

1. Start the level with an unbalanced 220 ohm right top resistor. Power on briefly and confirm the LCD measures approximately 0.47 V between midpoints; level must not complete.

2. Turn off power. Disconnect wires from the 220 ohm part, recycle or move it using the shelf rules, install the correct 330 ohm resistor and reconnect.

3. Power on. Confirm left and right midpoint voltages are both approximately 2.94 V, branch currents are near 6.25 mA, source current is near 12.5 mA, and LCD differential voltage is near zero.

4. Open the bridge component context menu: **Project Spark / Synchronize AND REPORT**. Check all 4 components and all 8 terminal aliases resolve.

5. Check [LEVEL TARGET RESULT] diagnostics: all five entries should be Passed=True; the level should display TARGETS 5/5 and Completed.

6. Test incorrect 220 ohm resistor, disconnected leg, direct midpoint short, source short, and reversed measurement leads. Check that faults are rejected without destabilizing previous levels.

8. Common blockers

Symptom	Cause / correction
Targets 1/5; only source current passes	Workstation runtime IDs not bound. Correct slot ID, smart/legacy override, bridge declaration and terminal suffix matching.
CurrentSlot differs from GameObject parent	Fix shelf initial registration: SlotId uniqueness, Occupant and CurrentSlot must agree.
Bridge voltage reads 0 on an open circuit	Zero alone is insufficient: require both branch-current targets and source-current target.
Wrong resistor still passes	Check upper-right current target and source-current range; exact ohms require an additional resistance-specific target if mandatory.
Only electrical results, not meter use	Current 5 targets verify solved states, not probe interaction; separate gameplay condition needed to require actual meter use.

9. Before marking Level 20 complete

Check: no duplicate SlotIds, declared IDs match active binding rules, all three sides of each physical circuit contact are wired correctly, no conductive wire between MID A and MID B, and the TXT import was applied to the ORIGINAL Level_20.asset (not an imported duplicate).

This guide specifies the intended design and theoretical expected values; final behavior must be verified in your running Project Spark solver. It does not change Levels 1-19 or your evaluator code.